Data dictionary

K Dataset

This dictionary describes the three CSV files in the K Dataset folder. The first two files are the source tables. The third file joins them. Nothing in the source files was changed before the join.

Files

- data centers.csv. 4,802 data-center facilities from OpenStreetMap, one row per facility. This is a copy of dataset/data_centers.csv.
- ember carbon intensity.csv. 194 countries. Carbon intensity of electricity generation in 2024, from Ember's yearly electricity data. One row per country.
- merged energy datasets.csv. The facility file with the 2024 country carbon figures attached. 4,802 rows. 4,786 facilities matched a country. 16 did not.

How the files connect

The join key is country_iso3, the three-letter ISO country code. The merge is a left join: every facility is kept. Ember columns are blank when that country has no 2024 total-generation intensity. Countries with no match are Ukraine, Lesotho, Isle of Man, Curacao, Faroe Islands, Tuvalu, and Kosovo, plus two facilities with no country.

Ember source: yearly electricity data, area type "Country or economy," electricity source "Total generation," year 2024. Published at ember-energy.org/data/yearly-electricity-data. The intensity figure is for all electricity generated in the country, not for the data center itself.

data centers.csv

One OpenStreetMap feature tagged as a data center or similar telecom building. About two-thirds of rows have facility_kind unclassified. Footprint is missing for about 19% of rows. Operator is missing for about 40%. Start date is missing for about 96%.

Column	Meaning
osm_type	OpenStreetMap element type: node (a point), way (an outline), or relation (a grouped object).
osm_id	OpenStreetMap identifier. Unique together with osm_type. Not unique on its own.
name	Facility name from the OpenStreetMap name tag. Blank when the map object has no name.
name_en	English name, when the object has a name:en tag. Usually blank.
operator	Organization that operates the site, from the operator tag.
operator_short	Short form of the operator name, when tagged.
operator_wikidata	Wikidata id for the operator, such as Q12345. Useful for matching a company across spellings.

Column	Meaning
owner	Owner, from the owner tag. Distinct from operator. Usually blank.
brand	Brand tag. Usually blank.
facility_kind	Type assigned to the site. Values: unclassified, colocation, hyperscale, telecom_carrier, research_or_academic, financial, government_or_public. Unclassified means the tags were not enough to choose a type.
tagging_scheme	Which OpenStreetMap tags identified the object: telecom, building, building+telecom, man_made, or building+man_made+telecom.
country	Country name assigned to the facility.
country_iso2	Two-letter ISO country code, such as US.
country_iso3	Three-letter ISO country code, such as USA. This is the join key to the Ember file.
continent	Continent of the assigned country.
subregion	UN subregion of the assigned country, such as Northern America or Western Europe.
country_match	How the country was chosen. boundary means the facility sits inside a country outline. nearest_Xkm means it was assigned to the nearest country, X kilometers away. Blank for two rows.
state	State, province, or region from the address tags. Missing for most rows.
city	City from the address tags. Missing for most rows.
postcode	Postal code from the address tags.
street	Street from the address tags.
latitude	Latitude in decimal degrees, WGS84.
longitude	Longitude in decimal degrees, WGS84.
footprint_area_m2	Ground area of the mapped outline, in square meters. Blank for node features and for ways with no measured outline. This is floor area of the map polygon, not power use.
building_levels	Number of floors, when tagged. Usually blank.
height_m	Building height in meters, when tagged. Usually blank.
start_date	Opening date or year from the start_date tag. Usually blank, so this file cannot show when most sites were built.
website	Website from the website tag.
wikidata	Wikidata id for the facility itself, not the operator.
wikipedia	Wikipedia page title for the facility, when tagged.

Column	Meaning
osm_last_edited	Date the OpenStreetMap object was last edited, YYYY-MM-DD.
osm_url	Link to the object on openstreetmap.org.

ember carbon intensity.csv

One row per country for 2024. Intensity is the emissions of the country's total electricity generation divided by the electricity generated. A lower number means a cleaner grid. It is a national average: every facility in a country gets the same value. It does not vary by US state or grid region.

Column	Meaning
country	Country name as Ember spells it.
country_iso3	Three-letter ISO country code. Join key to data centers.csv.
year	2024 for every row. 2025 exists in Ember's full release but covers fewer countries, so 2024 is the year used here.
emissions_intensity_gco2e_per_kwh	Grams of carbon-dioxide equivalent emitted per kilowatt-hour of electricity generated. Ember's field is "Emissions intensity (gCO2e/kWh)" on the Total generation row. Example: United States is 383.81.
generation_twh	Total electricity generated in that country in 2024, in terawatt-hours.
emissions_mtco2e	Power-sector emissions from that generation, in million tonnes of carbon-dioxide equivalent.

merged energy datasets.csv

Every column from data centers.csv, plus the four Ember columns below. Facility columns keep the definitions above. Ember values repeat for every facility in the same country, because the carbon figure is national.

Column	Meaning
year	Ember year attached to the facility. 2024 when the country matched. Blank when it did not.
emissions_intensity_gco2e_per_kwh	That country's 2024 grid carbon intensity, in gCO2e per kWh. Blank for the 16 unmatched facilities.
generation_twh	That country's 2024 electricity generation, in terawatt-hours. Same value for every facility in the country.
emissions_mtco2e	That country's 2024 power-sector emissions, in million tonnes of CO2e. Same value for every facility in the country.

What these columns do not measure

The files do not contain a facility's own electricity use, power capacity in megawatts, power usage effectiveness (PUE), electricity price, or cost. footprint_area_m2 is a size measure only. emissions_intensity_gco2e_per_kwh describes the national grid the facility sits in, not the emissions of that building.